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(54) **SLOTTED MEMS FORCE SENSOR**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 221 days.

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CPC **G01L 1/18** (2013.01); **B81B 3/0072** (2013.01); **B81B 2201/0264** (2013.01); **B81B 2203/04** (2013.01)

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CPC B81B 3/0072; G01L 1/18; G01L 1/20
See application file for complete search history.

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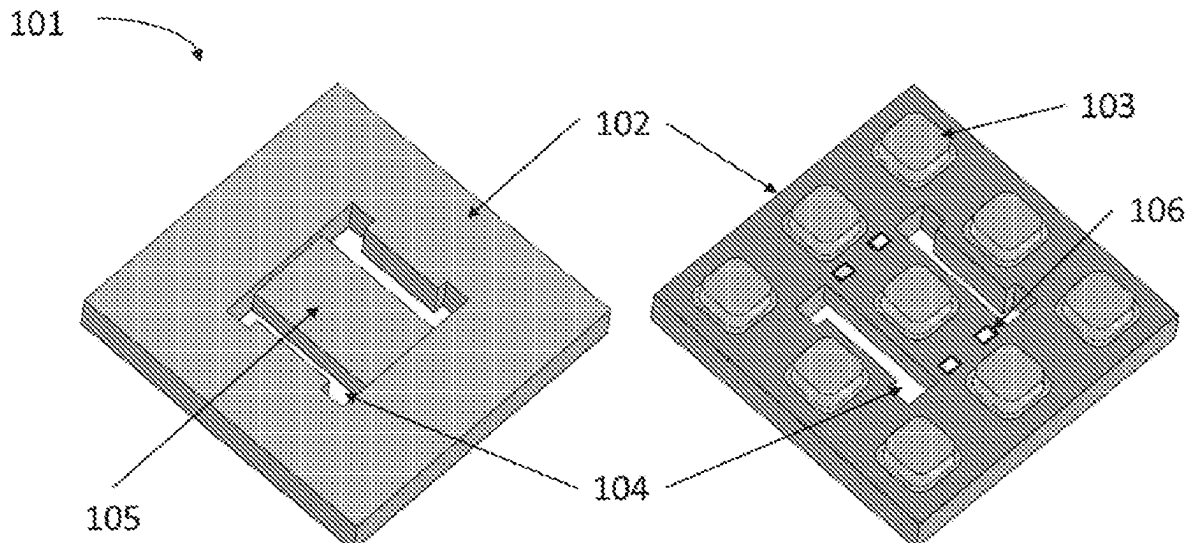
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(57) **ABSTRACT**

Described herein is a MEMS force sensor with stress concentration design. The stress concentration can be performed by providing slots, whether through or blind, and/or selective thinning of the substrate. The MEMS force sensor is in chip scale package with solder bumps or metal pillars and there are sensing elements formed on the sensor substrate at the stress concentrate area. The stress concentration can be realized through slots, selective thinning and a combination of both.

17 Claims, 5 Drawing Sheets



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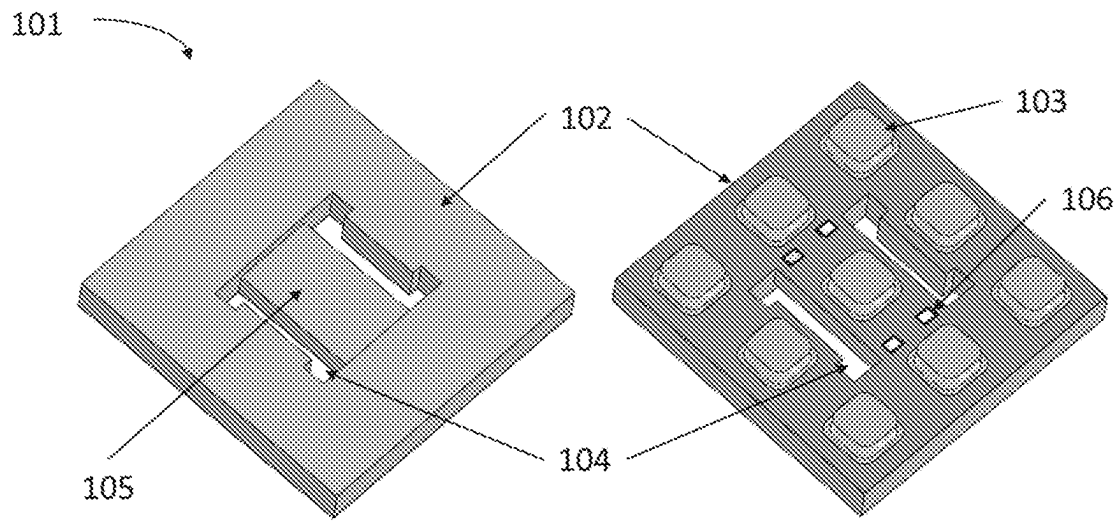


FIGURE 1

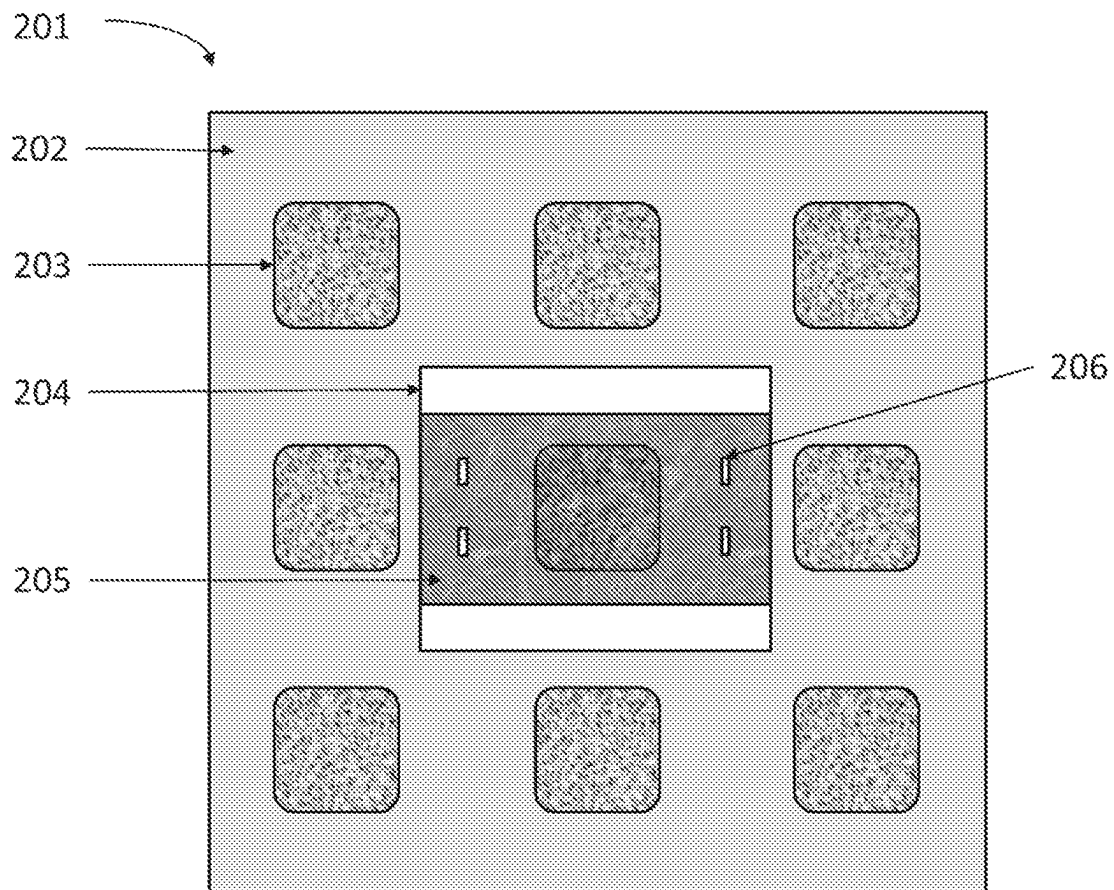
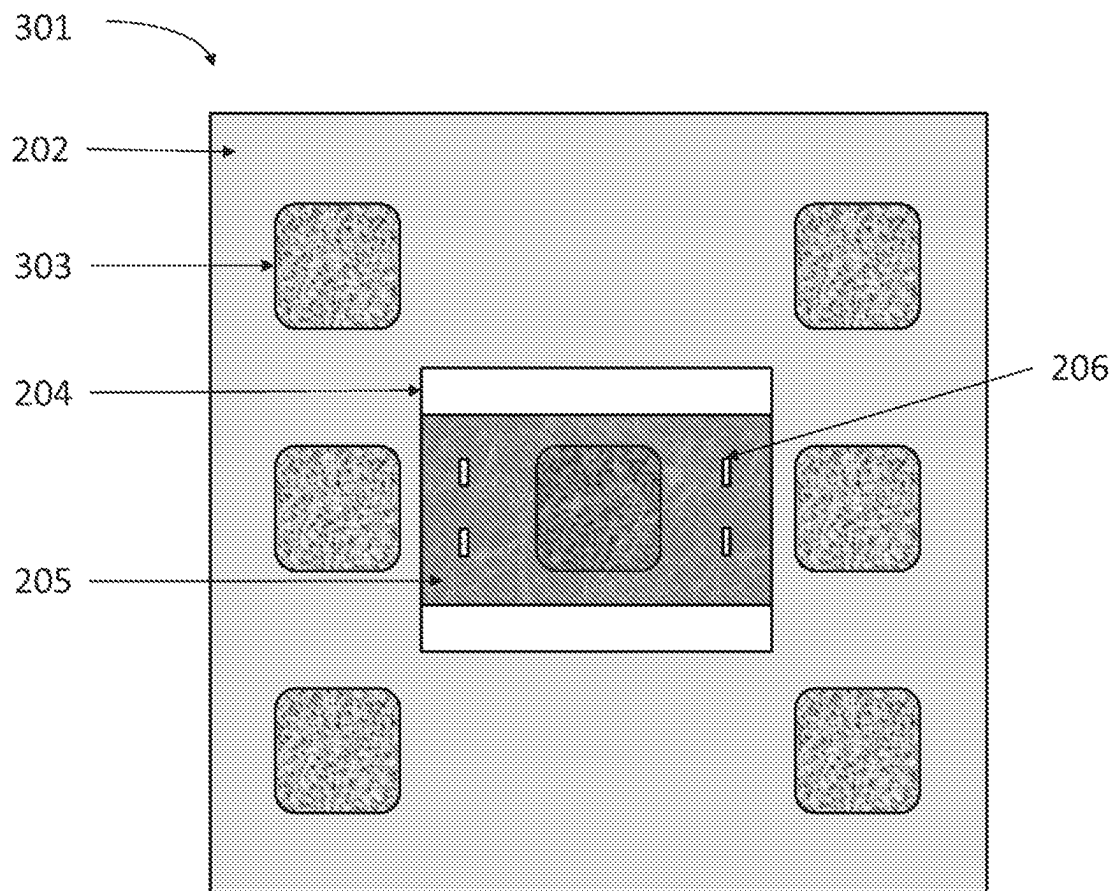


FIGURE 2

**FIGURE 3**

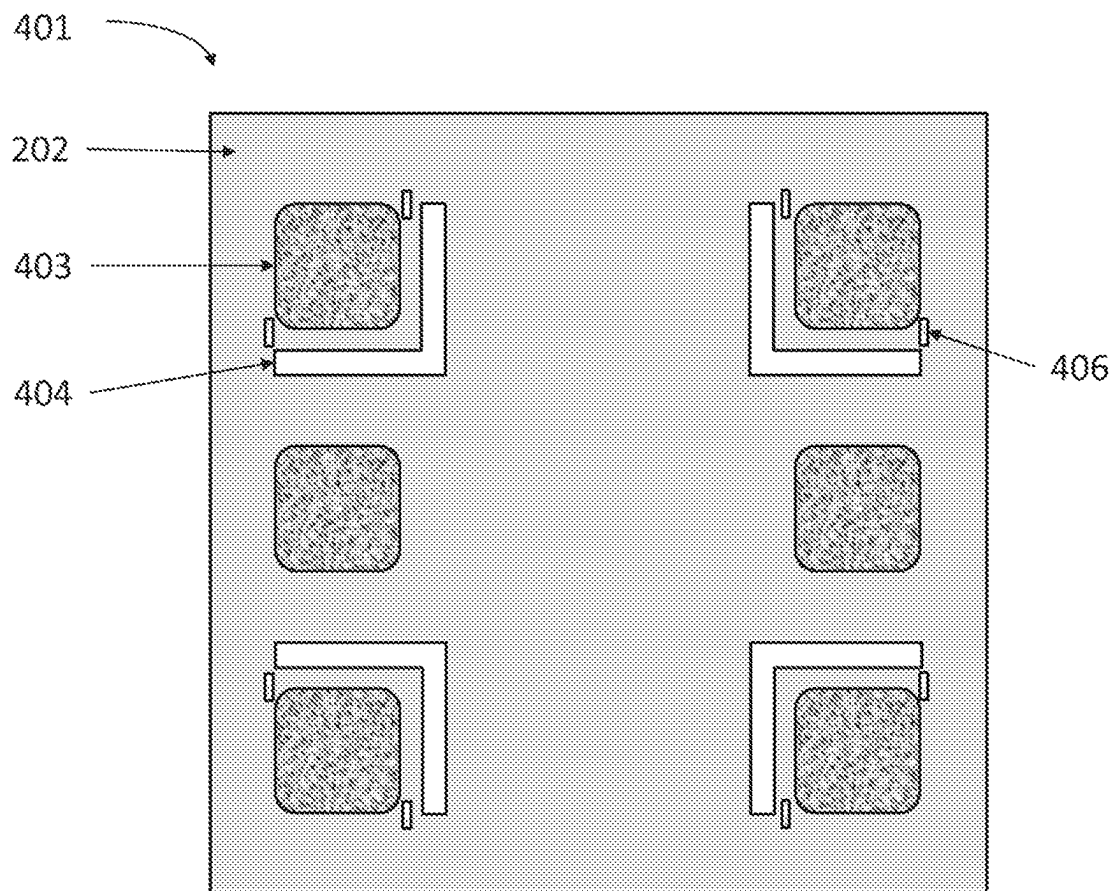


FIGURE 4

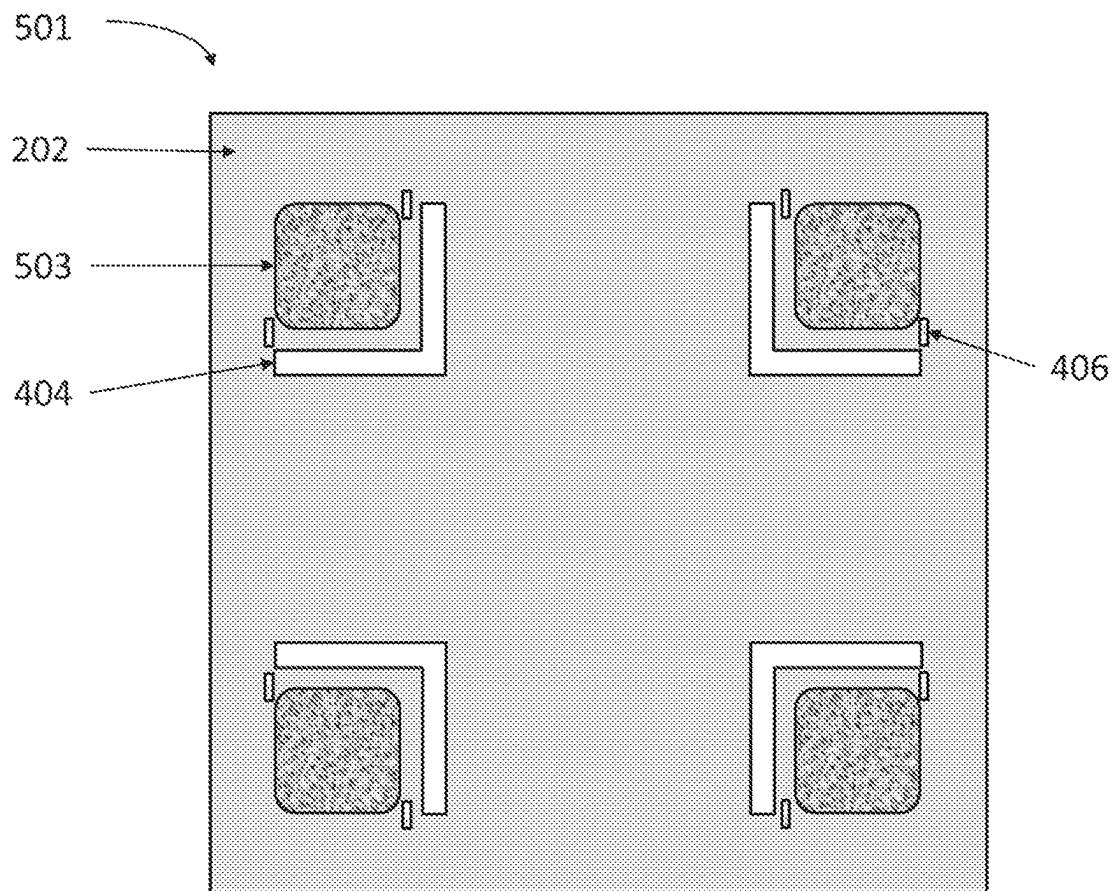


FIGURE 5

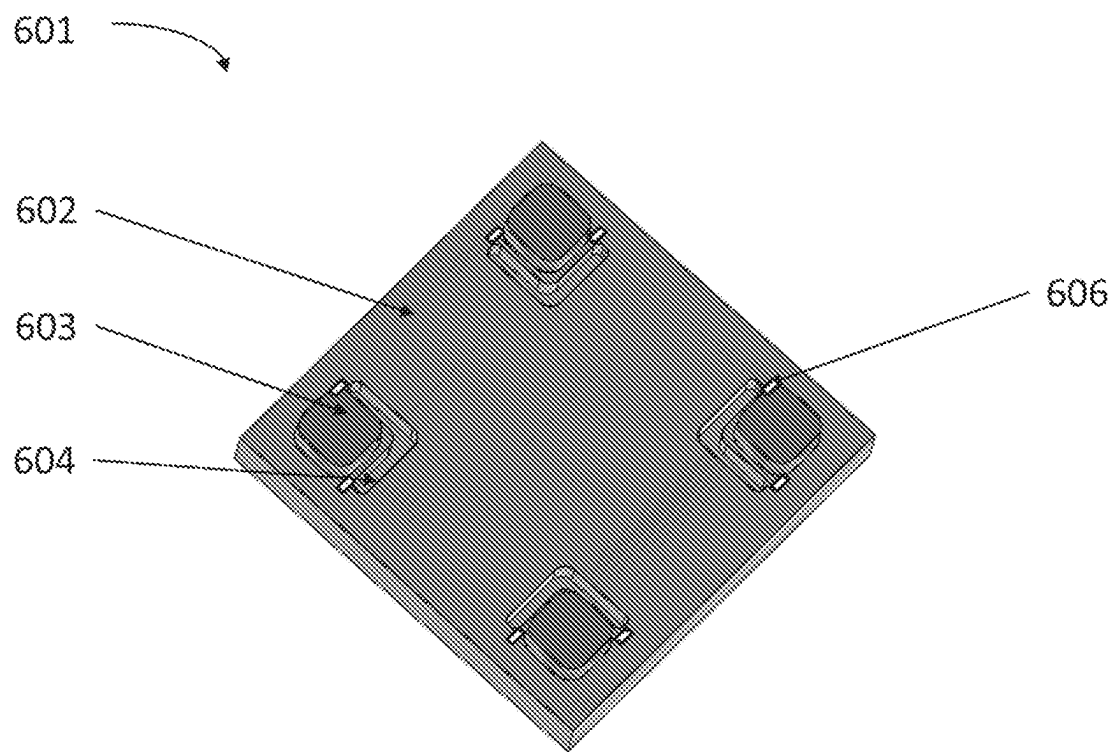


FIGURE 6

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SLOTTED MEMS FORCE SENSOR**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 17/215,186 filed on Mar. 29, 2021, now U.S. Pat. No. 11,698,310, which is a continuation of U.S. application Ser. No. 16/739,687 filed on Jan. 10, 2020, now U.S. Pat. No. 10,962,427, which claims the benefit of U.S. provisional patent application No. 62/790,513, filed on Jan. 10, 2019, and entitled "SLOTTED MEMS FORCE SENSOR," wherein each of the disclosures are expressly incorporated herein by reference in its entirety.

FIELD OF TECHNOLOGY

The present disclosure relates to a stress concentration design for a MEMS force sensor which improves the sensitivity.

BACKGROUND

The state of the art of MEMS force sensor design are in two major categories. One is a seal cavity sensor with the stress concentration already built-in. The other is single piece of substrate relying on the solder bumps arrangement to provide stress concentration. The latter is very easy for mass production and the cost is low, however the sensitivity of such MEMS force sensor is not optimal. At some extreme cases, the sensitivity may not be sufficient to functionalize a thick sensing substrate or through very soft material such as foam.

SUMMARY

The present disclosure pertains a MEMS force sensor with stress concentration design to increase the sensitivity for extreme usage cases such as hard sensing substrate or through a soft foam. By selectively removing material on the silicon substrate of the sensor chip, the maximum stress can be improved by a factor of 2 or more. This results in a MEMS sensor chip with slots, selective thinning region or a combination of both. The MEMS sensors described herein are not sealed sensors, e.g., there is no cap bonded to the sensor substrate.

An example microelectromechanical system ("MEMS") force sensor is described herein. The MEMS force sensor can include a semiconductor substrate, a plurality of electrical connectors disposed on the semiconductor substrate, at least one through slot formed on the semiconductor substrate, and at least one sensing element formed on the semiconductor substrate. The at least one through slot is disposed in a stress concentration region and is configured to create stress concentration. Additionally, the at least one sensing element is disposed in the stress concentration region. The at least one sensing element is configured to change an electrical characteristic in response to stress.

Optionally, the MEMS sensor further includes a selective thinning region formed on the semiconductor substrate. The selective thinning region is disposed in the stress concentration region and is configured to further increase the stress concentration.

In some implementations, the at least one sensing element is formed by implantation or diffusion.

In some implementations, the semiconductor substrate comprises electrical routing, where at least one of the

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electrical connectors and the at least one sensing element is electrically coupled via the electrical routing.

In some implementations, the electrical connectors are solder bumps or metal pillars.

5 In some implementations, the semiconductor substrate is formed of silicon or gallium arsenide.

In some implementations, each of the electrical connectors is arranged in a respective corner of the semiconductor substrate.

10 In some implementations, the at least one through slot is disposed in proximity to one of the electrical connectors. Optionally, the at least one sensing element is arranged between the at least one through slot and the one of the electrical connectors.

15 Another example MEMS force sensor is described herein. The MEMS force sensor can include a semiconductor substrate, a plurality of electrical connectors disposed on the semiconductor substrate, a selective thinning region formed on the semiconductor substrate, and at least one sensing element formed on the semiconductor substrate. The selective thinning region is disposed in a stress concentration region and is configured to create stress concentration. Additionally, the at least one sensing element is disposed in the stress concentration region. The at least one sensing element is configured to change an electrical characteristic in response to stress.

20 Yet another example MEMS force sensor is described herein. The MEMS force sensor can include a semiconductor substrate, a plurality of electrical connectors disposed on the semiconductor substrate, at least one blind slot formed on the semiconductor substrate, and at least one sensing element formed on the semiconductor substrate. The at least one blind slot is disposed in a stress concentration region and is configured to create stress concentration. Additionally, the at least one sensing element is disposed in the stress concentration region. The at least one sensing element is configured to change an electrical characteristic in response to stress. The at least one blind slot extends only partially through the semiconductor substrate.

25 In some implementations, the at least one blind slot is disposed in proximity to one of the electrical connectors. Optionally, the at least one sensing element is arranged between the at least one blind slot and the one of the electrical connectors.

30 Other systems, methods, features and/or advantages will be or may become apparent to one with skill in the art upon examination of the following drawings and detailed description. It is intended that all such additional systems, methods, features and/or advantages be included within this description and be protected by the accompanying claims.

BRIEF DESCRIPTION OF THE FIGURES

35 The components in the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding parts throughout the several views. These and other features of will become more apparent in the detailed description in which reference is made to the appended drawings wherein:

FIG. 1 is a three-dimensional (3D) perspective view of an example MEMS force sensor with 9 bumps and through slots and selective thinning region.

FIG. 2 is a top view of another example MEMS force sensor with through slots and selective thinning region.

FIG. 3 is a top view of another example MEMS force sensor with through slots and selective thinning region.

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FIG. 4 is a top view of another example MEMS force sensor with through slots located in proximity to corner electrical connectors.

FIG. 5 is a top view of another example MEMS force sensor with through slots located in proximity to corner electrical connectors.

FIG. 6 is a 3D perspective view of another example MEMS force sensor with through blind slots located in proximity to corner electrical connectors.

DETAILED DESCRIPTION

The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and their previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

The following description is provided as an enabling teaching. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made, while still obtaining beneficial results. It will also be apparent that some of the desired benefits can be obtained by selecting some of the features without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations may be possible and can even be desirable in certain circumstances, and are contemplated by this disclosure. Thus, the following description is provided as illustrative of the principles and not in limitation thereof.

As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a sensing element” can include two or more such sensing element unless the context indicates otherwise.

The term “comprising” and variations thereof as used herein is used synonymously with the term “including” and variations thereof and are open, non-limiting terms.

Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance may or may not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

The present disclosure relates to a MEMS force sensor design with slots, selective thinning or a combination of both.

FIG. 1 illustrates the 3D perspective view of a MEMS force sensor 101 with through slots and selectively thinning according to one implementation described herein. The MEMS force sensor 101 is not a sealed sensor, e.g., there is no cap bonded to the semiconductor substrate. The MEMS force sensor 101 includes a semiconductor substrate 102, a plurality of electrical connectors 103, through slots 104, a

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selective thinning region 105, and a plurality of sensing elements 106. In some implementations, the semiconductor substrate 102 can be formed of silicon (Si) or gallium arsenide (GaAs). In some implementations, the semiconductor substrate 102 can be formed of another material that can be doped form a piezoresistive sensing element. In the examples described herein, the electrical connectors 103 are solder bumps. In FIG. 1, the MEMS force sensor 101 includes nine electrical connectors 103, which are arranged in three rows. There is an electrical connector 103 provided in proximity to each of the corners of the semiconductor substrate 102. It should be understood that the number, size, shape and/or arrangement of the electrical connectors 103 in FIG. 1 are provided only as an example. This disclosure contemplates providing a MEMS force sensors having different number, sized, shaped and/or arrangement of electrical connectors. Additionally, although solder bumps are provided as an example, this disclosure contemplates using metal pillars (e.g., copper, nickel, or other metal) instead of solder bumps with the implementations described herein. It should be understood that the solder bumps and metal pillars are only provided as examples and that other types of electrical connectors can be used with the implementations described herein.

Additionally, the through slots 104 are openings that extend entirely through the semiconductor substrate 102. It should be understood that the through slots 104 can be formed by completely removing a portion of the semiconductor substrate 102 (e.g., by etching). A through slot 104 is arranged in a stress concentration region, which is the location of maximum stress during normal operation of the MEMS force sensor 101. The through slot 104 is configured to concentrate the stress. It should be understood that the number, size, and/or shape of the through slots 104 in FIG. 1 are provided only as an example. This disclosure contemplates providing MEMS force sensors having different number, sized, and/or shaped through slots. The selective thinning region 105 is a region on the semiconductor substrate 102 where a portion of the material is removed. In other words, the thickness of the substrate in the selective thinning region 105 is less than the thickness of the substrate in other regions of the semiconductor substrate 102 (e.g., near the edges of the substrate). It should be understood that the selective thinning region 105 can be formed by partially removing a portion of the semiconductor substrate 102 (e.g., by etching). The selective thinning region 105 is arranged in a stress concentration region, which is the location of maximum stress during normal operation of the MEMS force sensor 101. The selective thinning region 105 is configured to concentrate the stress. It should be understood that the number, size, and/or shape of the selective thinning region 105 in FIG. 1 are provided only as an example. This disclosure contemplates providing MEMS force sensors having different number, sized, and/or shaped selective thinning regions. As described herein, the MEMS force sensor can include a through slot, a selective thinning region, or both. When used in combination, the through slot and selective thinning region work together to concentrate the stress.

As discussed above, the MEMS force sensor 101 includes a plurality of sensing elements 106. As shown in FIG. 1, the sensing elements 106 are disposed on a surface of the semiconductor substrate 102. Optionally, in some implementations, the MEMS force sensor 101 can include a single sensing element 106. This disclosure contemplates that the sensing elements 106 can be diffused, deposited, or implanted on a surface of the semiconductor substrate 102.

The sensing elements **106** change an electrical characteristic (e.g., resistance, capacitance, charge, etc.) in response to deflection of the semiconductor substrate **102**. In one implementation, the sensing elements **106** are optionally piezoresistive sensing elements (e.g., piezoresistive transducers). Although piezoresistive transducers are provided as an example sensing element, this disclosure contemplates that the sensing elements **106** can be any sensor element configured to change at least one electrical characteristic (e.g., resistance, charge, capacitance, etc.) based on an amount or magnitude of an applied force and can output a signal proportional to the amount or magnitude of the applied force. Other types of sensing elements include, but not limited to, piezoelectric or capacitive sensors.

The sensing elements **106** are electrically coupled to the electrical connectors **103** through electrical routing (not shown) arranged on the semiconductor substrate **102** thus transforming the stress into electrical signal. The sensing elements **106** are strategically placed at a location where the stress is maximum (i.e., a “stress concentration region”) during normal operation. It should be understood that the stress concentration region(s) can be determined or located by simulation (e.g., finite element analysis) during design of the MEMS force sensor **101**. As described above, the through slots **104** and selective thinning region **105** are configured to concentrate stress in this region. In FIG. 1, the combination of the through slots **104** and the selective thinning region **105** together forms a thinned beam. This increases the sensitivity of the MEMS force sensor **101**. The maximum stress locations (i.e., the “stress concentration regions”) are typically at the edge of the thinned beam, which is the case for the MEMS force sensor **101**. As shown in FIG. 1, the sensing elements **106** are arranged on the thinned beam, e.g., on the selective thinned region **105**. Additionally, the sensing elements **106** are arranged in proximity to the edge of the thinned beam. In FIG. 1, the thinned beam forms a deformable membrane, which can be used for sensing strain. The sensing elements **106** are therefore configured to convert a strain on a surface of the semiconductor substrate **102** to an analog electrical signal that is proportional to the strain. It should be understood that the number and/or arrangement of the sensing elements **106** in FIG. 1 are provided only as an example. This disclosure contemplates providing a MEMS force sensors having different number and/or arrangements of sensing elements.

FIG. 2 further illustrates the two-dimensional (2D) view of a MEMS force sensor **201** with slots and selectively thinning according to another implementation described herein. The MEMS force sensor **201** is not a sealed sensor, e.g., there is no cap bonded to the semiconductor substrate. The MEMS force sensor **201** includes a semiconductor substrate **202**, a plurality of electrical connectors **203**, through slots **204**, a selective thinning region **205**, and a plurality of sensing elements **206**. The semiconductor substrate, electrical connectors, through slots, selective thinning region, and sensing elements are described in detail above with regard to FIG. 1. It should be understood that the semiconductor substrate **202**, electrical connectors **203**, through slot **204**, selective thinning region **205**, and sensing elements **206** in FIG. 2 may have characteristics as described above with regard to FIG. 1. The sensing elements **206** are electrically coupled to the electrical connectors **203** through electrical routing (not shown) arranged on the semiconductor substrate **202** thus transforming the stress into electrical signal. The sensing elements **206** are strategically placed at a location where the stress is maximum (i.e., a “stress concentration region”) during normal operation. As

described above, the stress concentration region(s) can be determined or located by simulation during design of the MEMS force sensor **201**. As described above, the through slots **204** and selective thinning region **205** are configured to concentrate stress in this region. Similar to FIG. 1, the combination of the through slots **204** and the selective thinning region **205** together form a thinned beam. This increases the sensitivity of the MEMS force sensor **201**. As shown in FIG. 2, the sensing elements **206** are arranged on the selective thinning region **205** and also arranged in proximity to the edge of the thinned beam. The maximum stress locations are typically at the edge of the thinned beam, which is the case for the MEMS force sensor **201**. In this implementation, there are a total of 9 electrical connectors **203**.

FIG. 3 illustrates another implementation similar to FIG. 2. The MEMS force sensor **301** is not a sealed sensor, e.g., there is no cap bonded to the semiconductor substrate. The MEMS force sensor **301** with slots and selectively thinning includes a semiconductor substrate **202**, a plurality of electrical connectors **303**, through slots **204**, a selective thinning region **205**, and a plurality of sensing elements **206**. The semiconductor substrate, electrical connectors, through slots, selective thinning region, and sensing elements are described in detail above with regard to FIG. 1. It should be understood that the semiconductor substrate **202**, electrical connectors **303**, through slot **204**, selective thinning region **205**, and sensing elements **206** in FIG. 3 may have characteristics as described above with regard to FIG. 1. The sensing elements **206** are electrically coupled to the electrical connectors **303** through electrical routing (not shown) arranged on the semiconductor substrate **202** thus transforming the stress into electrical signal. The sensing elements **206** are strategically placed at a location where the stress is maximum (i.e., a “stress concentration region”) during normal operation. As described above, the stress concentration region(s) can be determined or located by simulation during design of the MEMS force sensor **301**. As described above, the through slots **204** and selective thinning region **205** are configured to concentrate stress in this region. The combination of the through slots **204** and the selective thinning region **205** together forms a thinned beam. This increases the sensitivity of the MEMS force sensor **301**. The maximum stress locations are typically at the edge of the thinned beam, which is the case for the MEMS force sensor **301**. As shown in FIG. 3, the sensing elements **206** are arranged on the selective thinning region **205** and also are arranged in proximity to the edge of the thinned beam. In this implementation, there are a total of 7 solder bumps **303**.

FIG. 4 illustrates another implementation of the MEMS force sensor with through slots only (i.e., not including a selective thinning region). The MEMS force sensor **401** is not a sealed sensor, e.g., there is no cap bonded to the semiconductor substrate. The MEMS force sensor **401** includes a semiconductor substrate **202**, a plurality of electrical connectors **403**, through slots **404**, and a plurality of sensing elements **406**. The semiconductor substrate, electrical connectors, through slots, and sensing elements are described in detail above with regard to FIG. 1. It should be understood that the semiconductor substrate **202**, electrical connectors **403**, through slot **404**, and sensing elements **406** in FIG. 4 may have characteristics as described above with regard to FIG. 1. The through slots **404** are placed only in proximity to the electrical connectors **403** that are arranged at the corners of the MEMS force sensor **401**. This increases the sensitivity of the MEMS force sensor **401**. There are no through slots arranged near the other electrical connectors

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403 (e.g., middle connectors). The sensing elements 406 are electrically coupled to the electrical connectors 403 through electrical routing (not shown) arranged on the semiconductor substrate 202 thus transforming the stress into electrical signal. The sensing elements 406 are strategically placed at a location where the stress is maximum (i.e., a “stress concentration region”) which is located between the through slots 404 and electrical connectors 403. As shown in FIG. 4, a sensing element 406 is arranged in proximity to each opposing end of a through slot 404, which are “L” shaped. As described above, the stress concentration region(s) can be determined or located by simulation during design of the MEMS force sensor 401. As described above, the through slots 404 are configured to concentrate stress in this region. In this implementation, there are a total of 6 electrical connectors 403.

FIG. 5 illustrates yet another implementation of the MEMS force sensor with through slots only. The MEMS force sensor 501 is not a sealed sensor, e.g., there is no cap bonded to the semiconductor substrate. The MEMS force sensor 501 with slots includes a semiconductor substrate 202, a plurality of electrical connectors 503, through slots 404, and a plurality of sensing elements 406. The semiconductor substrate, electrical connectors, through slots, and sensing elements are described in detail above with regard to FIG. 1. It should be understood that the semiconductor substrate 202, electrical connectors 503, through slot 404, and sensing elements 406 in FIG. 5 may have characteristics as described above with regard to FIG. 1. The through slots 404 are placed only in proximity to the electrical connectors 403, each of which is arranged at a corner of the MEMS force sensor 501. This increases the sensitivity of the MEMS force sensor 501. There are no additional electrical connectors in the MEMS sensor 501 (compare to FIG. 4). The sensing elements 406 are electrically coupled to the electrical connectors 503 through electrical routing (not shown) arranged on the semiconductor substrate 202 thus transforming the stress into electrical signal. The sensing elements 406 are strategically placed at a location where the stress is maximum (i.e., a “stress concentration region”) which is between the through slots 404 and solder bumps 503. As shown in FIG. 5, a sensing element 406 is arranged in proximity to each opposing end of a through slot 404, which are “L” shaped. As described above, the stress concentration region(s) can be determined or located by simulation during design of the MEMS force sensor 501. As described above, the through slots 404 are configured to concentrate stress in this region. In this implementation, there are a total of 4 electrical connectors 503.

FIG. 6 further illustrates the 3D view of the similar implementation of the MEMS force sensor in FIG. 5. The MEMS force sensor 601 is not a sealed sensor, e.g., there is no cap bonded to the semiconductor substrate. The MEMS force sensor 601 with slots includes a semiconductor substrate 602, a plurality of electrical connectors 603, blind slots 604, and a plurality of sensing elements 606. The semiconductor substrate, electrical connectors, and sensing elements are described in detail above with regard to FIG. 1. It should be understood that the semiconductor substrate 602, electrical connectors 603, and sensing elements 606 in FIG. 6 may have characteristics as described above with regard to FIG. 1. The blind slots 604 are placed only in proximity to the solder bumps 603, each of which is arranged at a corner of the MEMS force sensor 601. This increases the sensitivity of the MEMS force sensor 601. The blind slots 604 do not extend entirely through the semiconductor substrate 602. This is in comparison to the through

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slots shown in FIG. 5. For example, the blind slots 604 are openings that extend only partially through the semiconductor substrate 602. It should be understood that the blind slots 604 can be formed by removing a portion of the semiconductor substrate 602 (e.g., by etching). It should be understood that the number, size, and/or shape of the blind slots 604 in FIG. 6 are provided only as an example. The sensing elements 606 are electrically coupled to the solder bumps 603 through electrical routing (not shown) arranged on the silicon substrate 602 thus transforming the stress into electrical signal. The sensing elements 606 are strategically placed at a location where the stress is maximum (i.e., a “stress concentration region”) which is between the blind slots 604 and solder bumps 603. As shown in FIG. 6, a sensing element 606 is arranged in proximity to each opposing end of a blind slot 604, which are “L” shaped. As described above, the stress concentration region can be determined or located by simulation during design of the MEMS force sensor 601. The blind slots 604 are configured to concentrate stress in this region. In this implementation, there are a total of 4 electrical connectors 603.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

The invention claimed is:

1. A microelectromechanical system (“MEMS”) force sensor, comprising:
 - a semiconductor substrate;
 - a selective thinning region formed on the semiconductor substrate;
 - at least one slot formed on the semiconductor substrate;
 - electrical connectors arranged on the semiconductor substrate;
 - at least one sensing element formed on the semiconductor substrate wherein:
 - the at least one sensing element is arranged on the selective thinning region;
 - the at least one sensing element is arranged between the at least one slot and one of the electrical connectors;
 - the at least one sensing element is electrically coupled to the electrical connectors.
2. The MEMS force sensor of claim 1, wherein the at least one through slot is configured to create stress concentration in a stress concentration region, wherein the selective thinning region is disposed in the stress concentration region and is configured to further increase the stress concentration.
3. The MEMS force sensor of claim 1, wherein the at least one sensing element is formed by implantation or diffusion.
4. The MEMS force sensor of claim 1, wherein the semiconductor substrate comprises electrical routing, and wherein the one of the electrical connectors and the at least one sensing element is electrically coupled via the electrical routing.
5. The MEMS force sensor of claim 1, wherein the electrical connectors comprises solder bumps or metal pillars.
6. The MEMS force sensor of claim 1, wherein the semiconductor substrate is formed of silicon or gallium arsenide.
7. The MEMS force sensor of claim 1, wherein each of the electrical connectors is arranged in a respective corner of the semiconductor substrate.

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8. The MEMS force sensor of claim 7, wherein each of the at least one through slot is disposed in proximity to a different one of the electrical connectors.

9. The MEMS force sensor of claim 8, wherein the at least one sensing element is arranged between the at least one through slot and the one of the electrical connectors.

10. A microelectromechanical system ("MEMS") force sensor, comprising:

a semiconductor substrate;

a selective thinning region formed on the semiconductor substrate;

at least one slot formed on the semiconductor substrate; electrical connectors arranged on the semiconductor substrate;

at least one sensing element formed on the semiconductor substrate wherein:

the at least one sensing element is arranged on the selective thinning region;

the at least one sensing elements is electrically coupled to the electrical connectors through electrical routing arranged on the semiconductor substrate to transform stress into electrical signal.

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11. The MEMS force sensor of claim 10, wherein the at least one sensing element is formed by implantation or diffusion.

12. The MEMS force sensor of claim 10, wherein the electrical routing comprises conductive traces.

13. The MEMS force sensor of claim 10, wherein the electrical connectors are solder bumps or metal pillars.

14. The MEMS force sensor of claim 10, wherein the semiconductor substrate is formed of silicon or gallium arsenide.

15. The MEMS force sensor of claim 10, wherein each of the electrical connectors is arranged in a respective corner of the semiconductor substrate.

16. The MEMS force sensor of claim 15, wherein each of the at least one through slot is disposed in proximity to a different one of the electrical connectors.

17. The MEMS force sensor of claim 16, wherein the at least one sensing element is arranged between the at least one through slot and the one of the electrical connectors.

* * * * *